

Business Analyst - Superstore Project

Prepared For: CEO, Marketing Team, Operations Team

Introduction

The strategic framework is designed to recover approximately \$150,000 in lost revenue by optimizing our seasonal return processes using Tableau for this analysis project. Based on an analysis of historical data from 2018 to 2021, I have identified key drivers of revenue leakage across our geographic, categorical, and logistical operations. Our focus will center on the critical Q3 and Q4 peak seasons—specifically August through December—where back-to-school and holiday surges currently coincide with high return rates. We will examine how logistical variables, customer expectations, and our current refund policy impact the bottom line, and how we can protect our competitive holiday return window while minimizing exploitation.

Executive Summary

- Tableau Dashboard and Storytelling Project: Strategies for reducing returned orders by analyzing the return rate (%)
- Planning to reduce 30% returning rate and increase about \$157,000 business revenue.
- Investigate the root causes by analyzing the return metric across geographic, categorical, and ship-mode dimensions during the peak-demand season.
- Recommendation to the Operation Team, Marketing Team for next steps, helping the business to optimize the operation and protect the company's return/ refund policy.

Methodology and Data Sources

1. Data Sources:

Data file: Superstore database (2018 - 2021)

2. Metric and Analytical Approach:

- Key Performance Indicators (KPIs): Track the Scaling Gap between order volume growth (43%) and the rise in return rates (24.5% to 29.3%) to ensure quality standards scale with sales.
- Return Rate (%) by Month: Calculate the potential for \$150,000+ revenue recovery by reducing return rates from 26% to a target of 17%.
- Return Rate (%) by Category: Analyze the impact of heavy goods (e.g., Chairs/Tables at 29.7%) on reverse-logistics costs. Evaluate high-volume/high-return sub-categories (e.g., Binders and Paper at ~28%) to identify potential vendor quality issues.

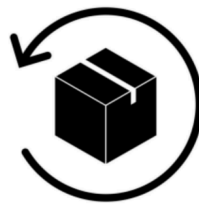
- Return Rate (%) of Sub-Category by Month: Correlate shipping speed with return frequency to determine if **expedited shipping** (e.g., First Class or Same Day) is compromising Quality Control (QC) standards.
- Return Rate (%) by State: Investigate "Negative Margin" states (e.g., OR, TN, CO, AZ) where high return rates cause processing fees to exceed product margins.

Analysis & Key Findings



Strategies To Reduce The Volumn of Retured Order By Analyze the Return Rate (%)

Data source: 2018 - 2021



- Lack of Control in Packing Process
- Lack of Control in Shipping Process
- Lack of Control in Quality of Products
- Winter - Snow/ Storm Season -> Products Damaged
- Customer Satisfaction for Product and Services
- Our Return/ Refund Policy

Business Analyst: Hai T Le

1. How has the revenue performance been over the 4 years (2018 -2021) and the goal of the analysis?

- **KPI Overview:** Focus on revenue, profit, order volume, and return rates.
- **Financial Growth:** Revenue scaled from **\$570K to \$1M (2018–2021)**, with a **300% surge in net profit (\$50K to \$150K)** signaling strong operating leverage.
- **Scaling Gap:** While order volume grew 43%, return rates rose from 24.5% to 29.3%, indicating fulfillment and quality standards are lagging behind sales.
- **Objective:** Identify root causes of return inflation to minimize reverse-logistics costs and maximize revenue retention.
- **Goal:** Implement a **strategy to reduce** the return rate **by 30%** (from **26%** down to **17%**).



Key Performance Indicators (KPIs)

Sales	Profit	Total Order	Total Return Order	Return Rate
\$2,901,676.77	\$360,129.51	12,420	3,226	25.97%

Sales and Profit



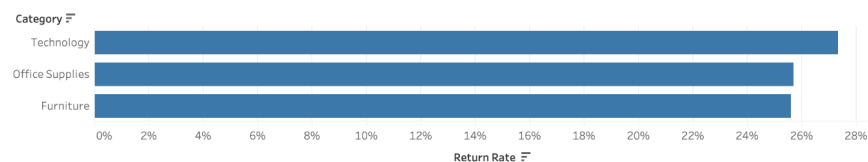
Orders and Return Rate



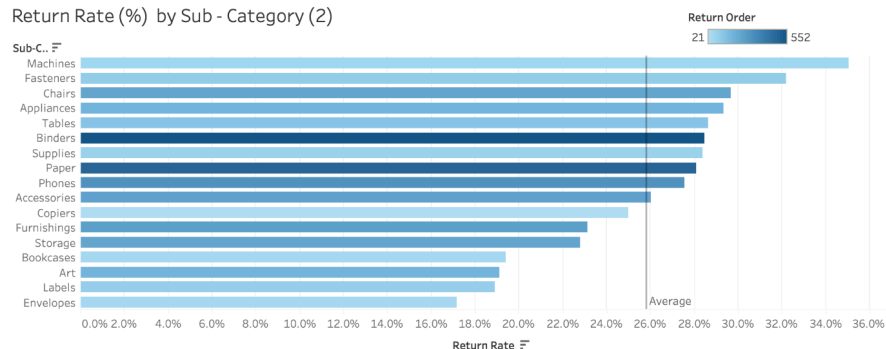
2. Return Rate by Category:

- **Segment Baseline:** Returns remain uniform across Technology, Furniture, and Office Supplies, indicating a systemic rather than segment-specific issue.
- **Technology Insights:** High-volume sub-categories like Phones (**27.6%**) and Accessories (**26%**) maintain healthier profiles than Machines, which show higher volatility.
- **Office Supplies Quality Risk:** Binders and Paper drive massive volume (3,600+ orders), but a **28% return rate** suggests a potential lapse in vendor quality or durability.
- **High-Logistical Risk (Furniture):** Chairs and Tables reach peak return rates of **29.7%**. Given the weight and dimensions of these goods, reverse logistics for these items represent the greatest threat to net margins.

Return Rate (%) by Category



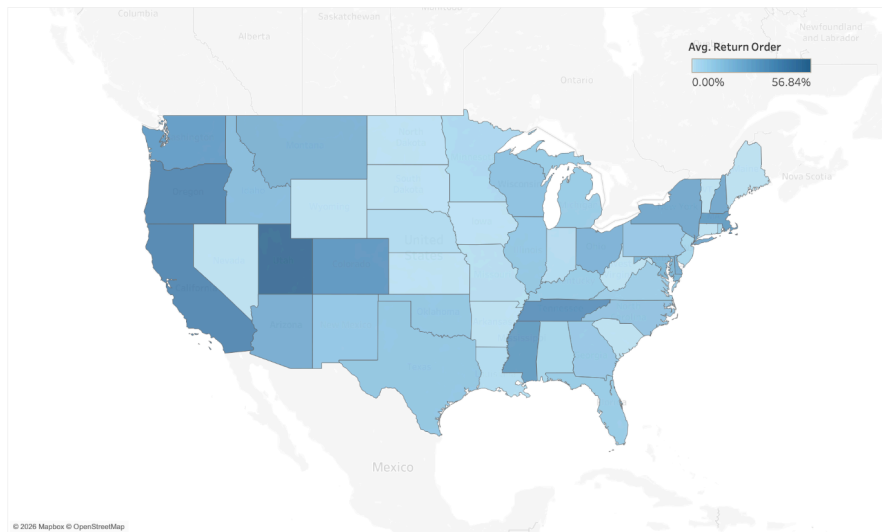
Return Rate (%) by Sub - Category (2)



3. Returned Rate by States:

- **Regional Variance:** Data visualization confirms that return volatility is non-uniform across the country, with the most significant concentration of revenue leakage isolated to the **West Region**.
- **Performance Benchmarks:** While several states have successfully maintained a **0% return rate**, these successes are offset by high-frequency return clusters in western territories.
- **Strategic Focus:** Future mitigation efforts and root-cause investigations will be prioritized in the West Region to stabilize fulfillment and quality standards.

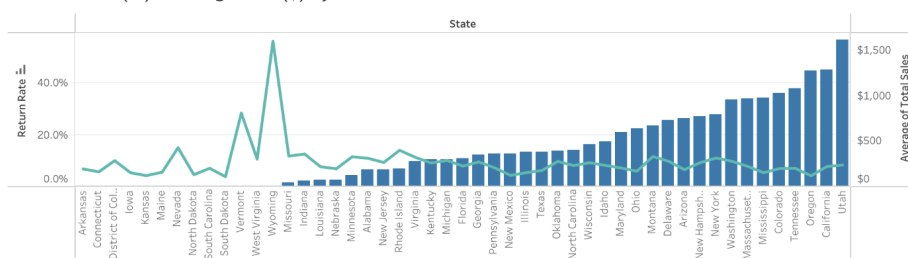
Return Rate (%) by State



4. Return Rate & Avg Sales, Return rate & Avg Profit:

- **Peak Season Volatility:** Analysis of the August–December window reveals that Utah and California exhibit the highest return concentrations during peak demand.
- **Margin Erosion:** A critical correlation exists in Oregon, Tennessee, Colorado, and Arizona, where high return rates have pushed net margins into the negative.
- **Profitability Leakage:** Evidence suggests that return processing fees in these four states are currently exceeding original product margins, necessitating a deep-dive investigation into localized fulfillment issues.

Return Rate (%) and Avg Sales (\$) by State



Return Rate (%) and Avg Profit (\$) by Month



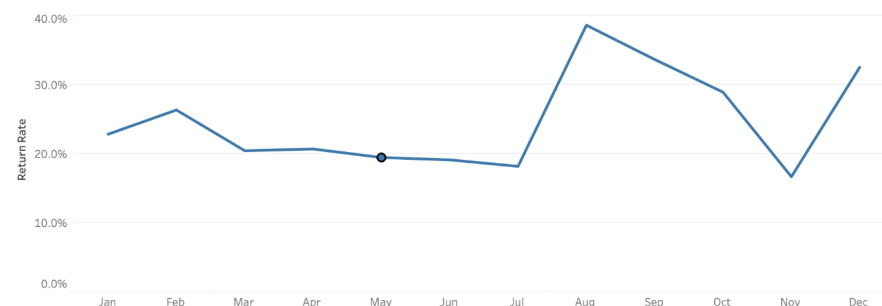
5. Ship Mode and Return Rate by month:

- **Regular Services:** Monitor Standard and Second Class shipping for seasonal spikes where return rates fluctuate between 15% and 36%.
- **Premium Service Vulnerability:** First Class return rates (19%–50%) highlight a systemic failure in maintaining QC under tight fulfillment windows.
- **Immediate Concern:** The 'Same Day' service demonstrates unsustainable return levels (peaking at 70.4%) during high-demand months (Aug–Dec).
- **Investigation Required:** We must audit the root causes—ranging from packing errors and weather-related damage to policy exploitation—to mitigate the high overhead of premium shipping returns.

Return Rate (%) by Ship Mode

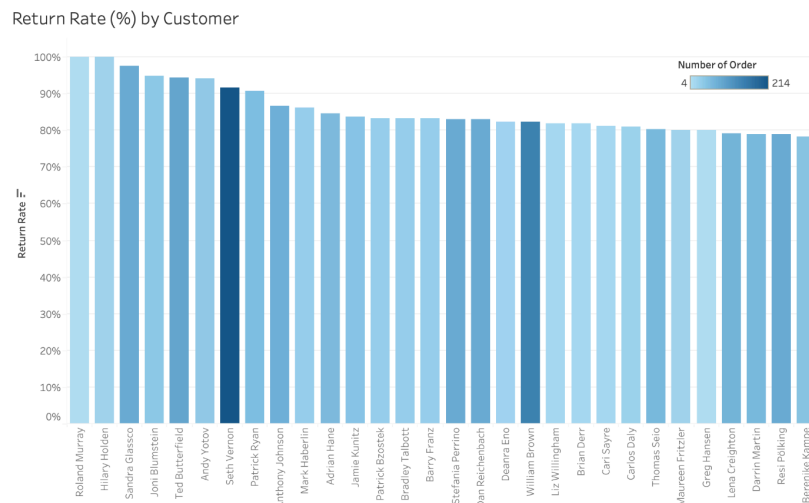
Ship Mode	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
First Class	0.0%	37.9%	44.0%	3.7%	19.1%	37.0%	18.6%	50.8%	20.4%	19.0%	26.1%	42.6%
Same Day	0.0%	0.0%	1.9%	0.0%	0.0%	0.0%	0.0%	53.7%	61.4%	63.6%	24.4%	70.4%
Second Class	13.8%	52.9%	7.6%	14.8%	3.5%	3.4%	11.0%	58.3%	14.0%	13.6%	12.9%	21.5%
Standard Class	28.8%	9.6%	18.3%	26.3%	24.4%	18.2%	22.0%	23.1%	36.8%	27.4%	14.8%	26.2%

Return Rate (%) by Month



6. Return Rate by Customer:

- **Return Rate Outliers:** Two high-volume accounts—Seth Vernon (94% returns) and William Brown (82% returns)—significantly skew regional metrics and suggest policy abuse.
- **Operational Intervention:** Recommend transitioning this "High-Return" segment (78%–100%) to a white-glove monitoring process.
- **Quality Assurance Action:** The Customer Service team will prioritize these accounts for order verification and satisfaction surveys to determine if returns are driven by product dissatisfaction or systematic abuse.



Recommendations and Conclusion:

Finance & Policy:

- **Tiered Service Fees:** Implement return shipping fees specifically for **Same Day** and **First Class** services to offset high reverse-logistics overhead.
- **Loyalty Differentiation:** Introduce specialized return windows for **loyal customers and paid members** to maintain a competitive advantage.
- **Store Credit Initiative:** Shift to **"Store Credit Only"** refunds for non-members to improve revenue retention and encourage future sales.

Operations & Logistics:

- **Shipping & Fulfillment:** Conduct a comprehensive contract review with shipping partners and heighten monitoring of the fulfillment lifecycle.
- **Quality Assurance:** Standardize specialized packing protocols for fragile and high-value goods to minimize transit damage.

Marketing & Customer Success:

- Execute deep-dive surveys on "problem products" and high-frequency returners.
- **Account Monitoring:** Launch a "High-Return" customer success program to investigate the satisfaction and behavior of the 78%+ return segment.



Conclusions

- Review our Return and Refund policy
- Review contracts with shipping companies
- Monitoring the shipping process
- Control the shipping process specially for easy damaged products.
- Take surveys for highest return rate products to find the main reasons
- Take surveys and monitor the highest return rate customers to find the reasons



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Tableau Dashboard: [Superstore Project](#)

